MILITARY SPECIFICATION

GRENADES, GP: M42 AND M46, INERT LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and provides quality assurance provisions for loading and assembling of two types of grenades designated as Grenades, GP: M42 and M46, Inert.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1235 - Single and Multi-Level Continuous Sampling Procedures and Tables For Inspection by Attributes

FSC 1330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9381451 - Grenade, GP: M42, Inert
9381454 - Grenade, GP: M46, Inert

PRODUCT AND PACKAGING DRAWINGS

9349337 - Packing and Marking for Box Outer for Grenade: M42, M46 and XM77

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Grenades. The grenades shall comply with all requirements specified on Dwg. 9381451 or 9381454, all associated drawings and with all requirements specified in applicable specifications.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:31Z
Check the source to verify that this is the current version before use.

3.3 Arming test of fuze slide of grenade. The grenade fuze, less safety clip and safety pin, shall have the arming screw rotated, and held out of the slide permitting the slide assembly to move to the armed position. The armed position shall be considered to have occured when the distance from the end of the tunnel of the fuze housing to the crown of the beveled end of the slide assembly measures .410 + .035 inch.

3.4 Grenade storage. Grenade bodies removed from the barrier bag packaging and stored as metal parts or partially or fully loaded assemblies, shall not be subjected to a temperature below 50°F (see 6.7).

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Records of results of contractor-conducted examinations and tests. The contractor shall maintain records of the results of all examinations or tests required by this specification. (See 6.4).

3.7 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. After loading, all external surfaces, including cavity, shall be free from inert filler. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawings	Quantity
Inert Grenades	9381451 or 9381454	100
Body Loading Assemblies	9381453 or 9381456	54
Packable Slider Lock	9329680	30

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

TABLE 1 - First Aritcle Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-70477 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Grenade and Subassembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inert Grenade (Dwg. 9381451 or 9381454)				
	Examination for defects (except 108-111)	100		3.2	4.4.2.6
	Unthreading torque test of clinched arming screw	100	-	3.2	4.5.3.2
	Arming test of fuze slide	100	-	3.2	4.5.3.3
	Drop load and torque test of clinched arming screw	100	-	3.2	4.5.3.4
	Pull test of clinched studs	100	-	3.2	4.5.3.5
	Body Loading Assy (Dwg. 9381453 or 9381456				
	Examination for defects (except 101)	54	-	3.2	4.4.2.5
	Consolidation density and weight of inert filler	4	-	3.2	4.5.2
	Unseating force of cone swage	50	-	3.2	4.5.4
	Packable Slider Lock (Dwg. 9329680)				
	Examination for defects	30		3.2	4.4.2.2

NOTES:

DPSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of grenades shall contain:

a. Fuzes, M223 of one lot interfix number from one manufacturer.
b. Body assemblies (Dwg. 9215343 or 9215373) of one lot interfix number from one manufacturer.
c. Cones of one lot interfix number from one manufacturer.
d. Tape-stiffener assemblies of one lot interfix number from one manufacturer.
e. Inert filler material of one lot interfix number from one manufacturer and no more than two serial lot numbers.
f. Lead cup from not more than two serial lots from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling in accordance with MIL-STD-1235 may be utilized if approved by the Procuring Activity.

b. Automatic gaging. Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with the contract.

c. Alternate inspection procedures. The contractor may propose the use of alternate inspection procedures in lieu of conducting sampling inspection for defects listed in this specification. (See paragraph 6.9).

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-70477 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215361
4.4.2.1	Tape Stiffener Assembly				NEXT HIGHER ASSEMBLY 9381454
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tear, twist or fold in tape		0.40%	3.2	Visual
102	Tape not completely inserted in slots in stiffener		0.40%	3.2	Visual
103	Crack in stiffener		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-70477 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Packable Slider Lock				9329680
					NEXT HIGHER ASSEMBLY 9381454
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile of minimum interior dimensions undersize		0.40%	3.2	Gage
102	Location of locking tab		0.40%	3.2	Gage
103	Length of locking tab		0.40%	3.2	Gage
104	Location of safety pin hole		0.40%	3.2	Gage
105	Width of safety pin hole		0.40%	3.2	Gage
106	Length of safety pin hole		0.40%	3.2	Gage
107	Pull test, max.		0.40%	3.2	4.5.5
108	Pull test, min.	(a)		3.2	4.5.5
109	Hardness test	(b or c)		3.2	4.5.6
110	Locking tab cracked, incomplete or missing		100%	3.2	Visual
Minor					
201	Width		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Protective coating incomplete or missing		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:
(a) The same sample from Major 107, accept on 0 - reject on 1.
(b) 25 samples per 4 hours production, accept on 0 - reject on 1, by the producer.
(c) When alternate material (Martensite steel) is used, apply special inspection level S-4, AQL 0.15%. Manufacturer's certification of hardness for preheat treated material is required.

DRDAR-QA Form 160a, 1 Jun 83

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-70477 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Body Assembly Type PH or CA Prior to Any Loading (see Note a)				9215343 or 9215373, as applicable
					NEXT HIGHER ASSEMBLY
					9381453 or 9381456
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any yellow marking present (applicable to type PH, Dwg. 9215343)		100%	3.2	Visual/Gage
102	Internal surface embossed (applicable to type CA, Dwg. 9215373)		100%	3.2	Visual/Gage
103	Verification of heat treatment (applicable to both types CA and PH)		100%	3.2	4.5.1
Minor					
201	Yellow marking incomplete or missing (applicable to type CA, Dwg. 9215373)		0.65%	3.2	Visual/Gage
202	Evidence of poor workmanship (applicable to both types CA and PH)		0.65%	3.7	Visual

NOTES: (a) These examinations are not required for metal parts originally produced on site and previously inspected for these defects by automatic gage.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil - Downloaded on 2026-01-23T03:37:03.0Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Fuze, M223, Prior to Clinching to Body Loading Assembly				9327872
					NEXT HIGHER ASSEMBLY
					9381452 or 9381455
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Inert marking incomplete, missing or illegible		100%	3.2	Visual
Major	None defined				
Minor	None defined				

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-70477 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9381453 or 9381456
4.4.2.5	Body Loading Assembly				NEXT HIGHER ASSEMBLY 9381452 or 9381455
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Inert marking incomplete, missing or illegible on body loading assembly		100%	3.2	Visual
2	Dot missing on empty lead cup		100%	3.2	Visual
Major					
101	Unseating force of cone swage	(a)		3.2	4.5.4
102	Evidence of any crack in body		100%	3.2	Visual/Gage (See 4.5.7)
103	Distance from open end to largest cone basic diameter		0.40%	3.2	Gage
104	Distance from largest cone basic diameter to smallest cone basic diameter		0.40%	3.2	Gage
105	Runout of cone at basic length		0.40%	3.2	Gage
106	Largest outside diameter, max.		0.40%	3.2	Gage
107	Lead cup missing or improperly assembled	(b)	0.40%	3.2	Visual/Gage
108	Cone swage missing, incomplete or improper		0.40%	3.2	Visual
109	Any part cracked, dented, distorted or missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:
(a) For each test select 50 samples per lot or 20 samples per machine per eight hours production, accept no failures reject on 1 failure.
(b) Inspection performed before loading inert filler.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.6	Grenade				9381451 or 9381454
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Inert marking incomplete, missing or illegible		100%	3.2	Visual
Major					
101	Arming screw not free to rotate		0.40%	3.2	Visual/Manual
102	Evidence of loose or incorrectly positioned tape		0.40%	3.2	Visual
103	Safety clip of fuze missing, damaged or double end not protruding beyond end of fuze tunnel		0.40%	3.2	Visual
104	Arming screw clinch missing, incomplete or not properly centered		0.40%	3.2	Visual
105	Fuze housing cracked or deformed		0.40%	3.2	Visual
106	Stud clinch cracked, split, missing, incomplete or not properly centered		0.40%	3.2	Visual
107	Washer missing		0.40%	3.2	Visual/Gage
108	Unthreading torque test of clinched arming screw	(a)		3.2	4.5.3.2
109	Arming test of fuze slide	(a)		3.4	4.5.3.3
110	Drop load and torque test of arming screw clinch	(a)		3.2	4.5.3.4
111	Pull test of clinched studs	(a)		3.2	4.5.3.5
112	Slide, protruding from fuze indicating armed or partially armed conditioned		100%	3.2	Visual

NOTES:
(a) 2 samples (See 4.5.3.1) per 1800 grenade assemblies produced on each machine, accept on 0 reject on 1

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-70477 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.6	Grenade				9381451 or 9381454
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Evidence of gap between fuze housing and cover or fuze cover and grenade cap or loose fuze stake		0.40%	3.2	Visual/Manual
114	Slider lock missing, cracked, incomplete or improperly positioned		100%	3.2	Visual
115	Height from the shoulder of grenade body to bottom of wound ribbon (two places)		0.40%	3.2	Gage
116	Presence of moly disulfide on phosphate coated slider lock		0.40%	3.2	Visual
117	Pull test, max.		0.40%	3.2	4.5.5
118	Pull test, min.	(b)		3.2	4.5.5
119	Safety pin of fuze missing or damaged		100%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

(b) The same samples from Major 117, accept on 0, reject on 1.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Packing and Marking for Box, Fiberboard for Grenade, GP: M42, M46 and XM77				9349336
					NEXT HIGHER ASSEMBLY
					9349337
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Partitions missing		0.40%	3.2	Visual
102	Tubes missing		0.40%	3.2	Visual
103	Barrier bag torn, punctured or heat seal incomplete		0.40%	3.2	Visual
104	Box damaged, exposing contents		0.40%	3.2	Visual
105	Corners not blunted		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Box marking missing, incomplete or illegible		0.65%	3.2	Visual
203	Barrier bag marking missing, incomplete or illegible		0.65%	3.2	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Packing and Marking for Box, Outer for Grenade: M42, M46 and XM77				9349337
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents loose		0.40%	3.2	Manual
102	Box damaged, exposing contents		0.40%	3.2	Visual
103	Strapping missing (wood box only)		0.40%	3.2	Visual
104	Steal missing (wire bound box only)		0.40%	3.2	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is required as described herein and in the First Article and Quality Conformance Inspection Table.

4.4.3.1 Consolidation density and weight of inert filler of body loading assembly (dwg. 9381453 or 9381456) Major defect. The density and weight of the inert material of the body loading assembly shall be determined on one assembly from each body loading machine at the start of production each day, and after every one hour of production or portion thereof. If the density or weight of any body loading assembly fails to comply with the drawing requirement, the body loading assemblies produced since the last satisfactory check shall be rejected. In addition, the necessary adjustments shall be made to the machine which will be verified by determining the consolidation density and weight of inert filler of two consecutively produced body loading assemblies. Testing shall be in accordance with paragraph 4.5.2.

4.4.4 Inspection equipment. For the inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, paragraph 6.3 and paragraph 6.8 herein.

4.5 Methods of inspection.

4.5.1 Verification of heat treatment of body assembly type PH or CA. This inspection shall employ the eddy current method, or any other equivalent method approved by the procurement agency. The processes, equipment and calibration procedures used in performing this test, and the hardness limits established shall be documented and have prior written approval of the procurement agency. In addition, after this approval has been granted, the contractor shall make no changes in equipment or procedures without prior written approval of the procurement agency. All equipment used shall be capable of performing this inspection accurately and consistently and shall be calibrated on an approved periodic basis. If the eddy current method is used, all personnel operating and calibrating this equipment shall be qualified and certified in accordance with MIL-STD-410. If any body assembly fails to comply with the requirements specified on the applicable drawing, it shall be classed defective and removed from production.

4.5.2 Consolidation density and weight of inert filler of body loading assembly. The following procedure shall be used to determine the weight and density of the inert filler of the body loading assembly. (When weighing in water, use a distilled water containing a suitable wetting agent such as "Aerosol" at 1cc per 250cc distilled water).

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:31Z
Check the source to verify that this is the current version before use.

a. Prior to assembly, weigh body assembly with lead cup in air (grams).
b. Prior to assembly, weigh the same body assembly with lead cup in water (grams).
c. Dry the components and load with inert filler using normal production methods.
d. Weigh the assembly in air (grams).
e. Weigh the assembly in water (grams).
f. Calculate the weight of inert filler in air (d) - (a).
g. Calculate the weight of inert filler in water (e) - (b).
h. Calculate weight (and volume) of water displaced by the inert filler (f) - (g).
i. Calculate the density of loaded inert filler (f) ÷ (g). grams/cc.

The body assemblies with lead cup inserted shall be randomly selected prior to loading and identified as test samples. Each sample shall be marked clearly with its weight and the applicable inert material consolidation machine number. The samples must be retrieved after powder consolidation but prior to cone insertion.

NOTE: Contractors utilizing the cone for re-consolidation of the inert filler shall develop a procedure for performing this test. The proposed procedure shall be submitted to the address specified in para. 6.4 for approval, prior to use.

4.5.3 Fuze to grenade assembly test

4.5.3.1 Preparation of sample items. The samples to be subjected to this inspection shall consist of a fuze, a tape stiffener assembly, and an empty body assembly from lots use in production of the grenade inspection lot. (See 4.4.1) They shall be prepared at the specified times from each assembly machine used during lot production. The samples shall be suitably marked to prevent mixing with grenades not intended to be samples, and to allow identification of the particular assembly machine on which the unit was produced.

4.5.3.2 Unthreading torque test. The sample shall be mounted in the approved fixture. The safety clip and the safety pin shall be removed. With the fixture preventing the fuze slide from arming, the torque required to rotate the arming screw out of the slide shall be measured. If the required torque exceeds the amount specified on the applicable drawing, the sample shall be classed defective.

4.5.3.3 Fuze slide arming test. After completion of the unthreading torque test, while maintaining the arming screw free of the slide, the fixture device for preventing slide movement shall then be withdrawn permitting the slide to move to the armed position. After observation for compliance with the arming

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:31Z
Check the source to verify that this is the current version before use.

requirements and while still maintaining the arming screw free of the slide, the slide shall be returned to the unarmed position. The safety pin and safety clip shall be properly reinserted in the fuze. Grenade fuzes failing to comply with the specified arming distance, shall be classed defective.

4.5.3.4 Drop load and torque test of arming screw clinch of grenade fuze. After completion of the arming test, the same sample shall be mounted in an approved fixture and subjected to the drop load as specified on the applicable drawing. The crimp of the tape stiffener assembly to the arming screw shall then be subjected to the torque as specified on the applicable drawings. If the tape rotates relative to the arming screw, when the specified torque is applied, the assembly shall be classed defective.

4.5.3.5 Pull test of clinched studs of body fuze assembly. After completion of the drop load and torque test of the arming screw clinch, the same sample assemblies (with safety clip and safety pin properly inserted in the fuze) shall have the tape stiffener assembly removed. This may be accomplished by any method which will not damage the arming screw threads. After removal of the tape stiffener assembly, the safety clip only shall be removed and the arming screw rotated to the armed position. The safety clip shall then be reinserted sufficiently to prevent the arming screw from freely rotating in the fuze while a suitable adapter is assembled to the threads on the protruding end of the arming screw. The sample item shall then be assembled to an approved fixture and the load specified on the applicable drawing shall be applied to the clinched studs by pulling on the assembled adapter. If the fuze loosens as evidenced by a gap between the body assembly and the fuze, the sample assembly shall be classed defective.

4.5.3.6 Disposition. If any body assembly fails to comply with any of these tests the production quantity represented by that sample shall be rejected.

4.5.4 Unseating force test of cone swage.

4.5.4.1 Preparation of sample items and test procedures for unseating force. The sample assemblies shall be randomly selected from normal production.

The quantity of body loading assemblies shall be cut in two in a plane perpendicular to the longitudinal axis of the body. The cut shall be located at a minimum distance of .700 inches from the open end. The cut shall be made cleanly through the body wall and explosive material.

The portion of the cut sample containing the swage of the cone shall have any remaining filler material cleaned away and then shall be mounted in an approved fixture. The load as specified on the

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T03:31Z
Check the source to verify that this is the currentversion before use.

applicable drawings shall be applied to the cone. If the cone is dislodged from its swaged position in the body loading assembly by the specified load, that body loading assembly shall be classed defective.

4.5.4.2 Disposition. If any body loading assembly fails to comply with the tests the production quantity represented by the test sample shall be rejected.

4.5.5 Pull test of the slider lock (non-destructive). Each sample slider lock shall be assembled to a fixture simulating a maximum dimension fuze housing and pull tested until the slider lock releases. The load shall be applied to the top of the lock in a direction away from the simulated housing. Any slider lock failing to meet the force requirement on the applicable drawing shall be classed defective. The lot represented by this sample shall be rejected.

4.5.6 Hardness test of the slider lock (non-destructive). This test shall be performed in accordance with ASTM-E-18 using the standard penetrator for the specified hardness scale. Readings shall be taken on the top surface of the slider lock. Any slider lock failing to meet the applicable hardness requirement shall be classed defective. The lot represented by this sample shall be rejected.

4.5.7 Inspection of body for cracks. This inspection shall be performed on each body after loading either visually or with an approved NDT method such as Eddy Current. If the eddy current method is used, all personnel operating and calibrating this equipment shall be qualified and certified in accordance with MIL-STD-410. Inspection shall be performed at each of the following locations.

a. Longitudinal defects in the wall of the body from the open end to the shoulder area.
b. Circumferential defects in the cap area of the body originating from the exterior surface.

4.5.7.1 Definition of crack. For the purpose of identifying a reject, the following definition of a rejectable crack will apply:

a. Crack - A discontinuity in the material with an orientation, depth, and length represented by approved standards.

5. PACKAGING. (This section is not applicable to intraplant usage).

5.1 Packaging. Level A (not intended for long term storage) and Level B: Packaging shall be in accordance with drawing 9349336.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

5.2 Packing. Level A (not intended for long term storage) and Level B: Packing shall be in accordance with drawing 9349337.

5.3 Marking. Marking of fiberboard box and bag shall be in accordance with drawing 9349336. Outer box shall be marked in accordance with drawing 9349337.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use with the Projectile, 155MM: Inert, M483A1.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to: Commander, ARDC, ATTN: AMSMC-QAR-I(D), Dover, New Jersey 07801-5001.

6.4 Data requirements. Data item descriptions required for submission under this specification are as follows:

Paragraph	Data requirement	Applicable D.I.
4.5.1, 4.5.3, 4.5.4, 4.5.5, 4.5.6,	Quality Inspection Defect Report	DI-R-1721
4.5.2	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724
6.9	Procedure Report (Tailored)	*DI-T-5372

*Delete a, b, c, d of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

The above data shall be forwarded in accordance with the requirements of the contract to Commander, ARDC, ATTN: AMSMC-QAR-Q(D), Dover, NJ 07801-5001.

6.5 Barricades. Proper barricade shall be utilized for all inspections performed, when applicable.

6.6 Drawings. Drawings listed in Section 2 of this specification under the headings US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal drawings. Technical data originally prepared by these activities or ARRADCOM is now under the cognizance of ARDC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

6.7 Grenade storage. The elapsed time between the loading and assembling of the grenades and their assembly into the projectile shall be kept to a minimum.

6.8 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to a periodic verification and a calibration check at frequent intervals during normal production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/ calibration. These procedures shall be included in the operation/ calibration procedures as required by paragraph 4.4.4.

6.9 Alternate inspection procedures.

6.9.1 Justification for alternate inspection procedures. Submitted documentation must include sufficient information to indicate an economic benefit, and must clearly identify that there is no reduction in product quality.

6.9.2 Alternate inspection procedures. The alternate inspection procedures as a minimum must include:

a. Frequency (production interval) for validating equipment functioning.
b. Proposed method of validation to include process and inspection drawings.
c. Description of material control procedures to be imposed during the production interval between validations (Note: Under these provisions final acceptance will be withheld from material produced during a production interval until conducting satisfactory validation.)

6.9.3 Approval of alternate inspection procedures. Proposed plans shall be submitted per para. 6.4 for approval.

6.10 Submission of data cards. In addition to the normal distribution of data cards, when the grenade is procured by the US Army Armament Material and Readiness Command, one (1) copy of all ammunition data cards shall be forwarded to: Commander, US Army ARDC, ATTN: AMSMC-QAR-Q(D), Dover, NJ 07801-5001.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1330-A212)

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:31Z
Check the source to verify that this is the currentversion before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT MUNITIONS & CHEMICAL COMMAND
ATTN: AMSMC-QA
DOVER, NJ 07801 5001

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-70477	GRENADES, GP: M42 AND M46, INERT LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:31Z
Check the source to verify that this is the current version before use.